Physics 4B Lab 5: *Ohm's Law, Part 1*

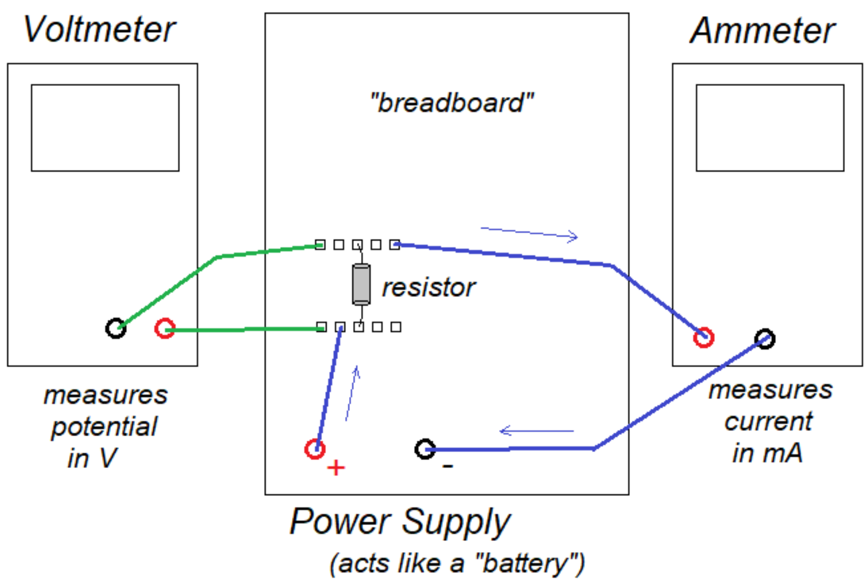
The objective of this lab is to use Ohm's Law to measure the resistance of three small resistors. For Part 2, we will consider how resistors behave when they are arranged in series *or in* parallel *in a circuit.*

Ohm's Law states that the electric current flowing through a *resistor* is proportional to the *potential difference* across the resistor and inversely proportional to the *resistance* of the resistor. This can be written as a simple equation, where I Is the electric current, V is the potential difference, and R is the resistance:

$$I = \frac{V}{R} \quad or \quad V = I\,R$$

If the potential is measured in *Volts*, and resistance is measured in *Ohms* (these are the SI units of potential and resistance), then the current is in units of *Ampere*, or "Amps".

We can measure the current through a resistor, with potential applied by a power supply, by use of an *ammeter*. Our apparatus includes a "breadboard" which we can use to connect a power supply, a resistor, and an ammeter to create a "circuit loop":



Note that current will flow through the "loop" of the circuit created by the wires shown in blue on the diagram. The wires shown in green allow us to connect a *voltmeter* to the circuit, which will measure the potential difference across the resistor (but current does not flow through these wires.)
A summary of the procedure:

- for each of three resistors, collect ten values of the potential and corresponding current
- create a data table of current in milliAmps and potential in Volts for each of the three resistors
- create a graph of V *vs.* I for each of the three resistors
- using Ohm's Law, $V = R\,I$ we should find the data forms a straight line with the slope of R

Part 1: Three Individual Resistors

We will use these resistors for Ohm's Law Part 2, as well as in a future lab. Identifying which resistor goes with which data will be very important. Be sure to record, in the data table, the colors of the three bands on your resistors. (We will also save your three resistors in a safe place so you can use the exact same resistors with the later lab.)

- For each of the three resistors, measure the current through the resistor for potential values of approximately 1, 2, 3… 10 Volts. **Be sure to record exactly what is on the measuring device**, i.e. do not round off your measurements.

- Create a graph of *V vs I* for each of the three resistors. ***Plot all three on one graph***. Include a "legend" to clearly label which plot belongs to which data set. I highly recommend using color to connect the labeling with the three individual plots.

- Format your equations for the best-fit lines to three decimal places. The slope of each data set is the value of the resistance, since ***V = R I*** and you have used "V "on your y-axis and "I" on your x-axis. But note that since the current is in *milliAmps*, i.e. 10^{-3} A, the units of the slope are

  Volt / 10^{-3} Amps 10^{3} Ohms or *kiloOhms*

  This means the resistance in Ohms is 1000 times the value of your slope. Record the values of the resistance, in Ohms, for each resistor in the small table below your main data table.

Sample graph for Part 1: